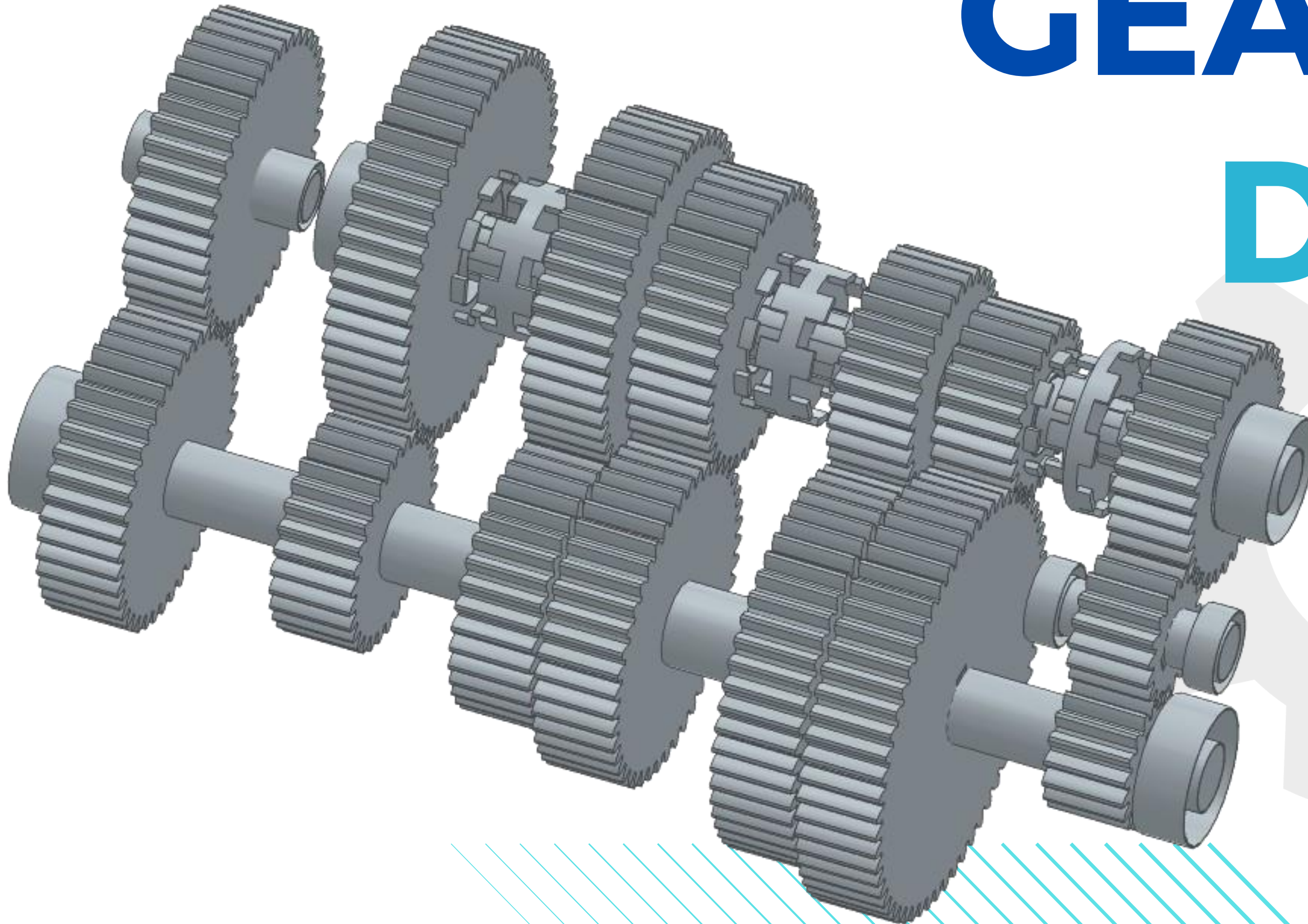


GEAR BOX DESIGN

Group 11

210660J
210640A
210494D



VEHICLE SELECTION

Given Design Parameters

- Weight of the vehicle = 2050 kg
- No. of passengers = 5
- Speed range = 0 - 170 kmph
- Speed at 300 inclined road = 35 kmph
- No. of forward gears = 5
- No. of reverse gears = 1

Selected vehicle

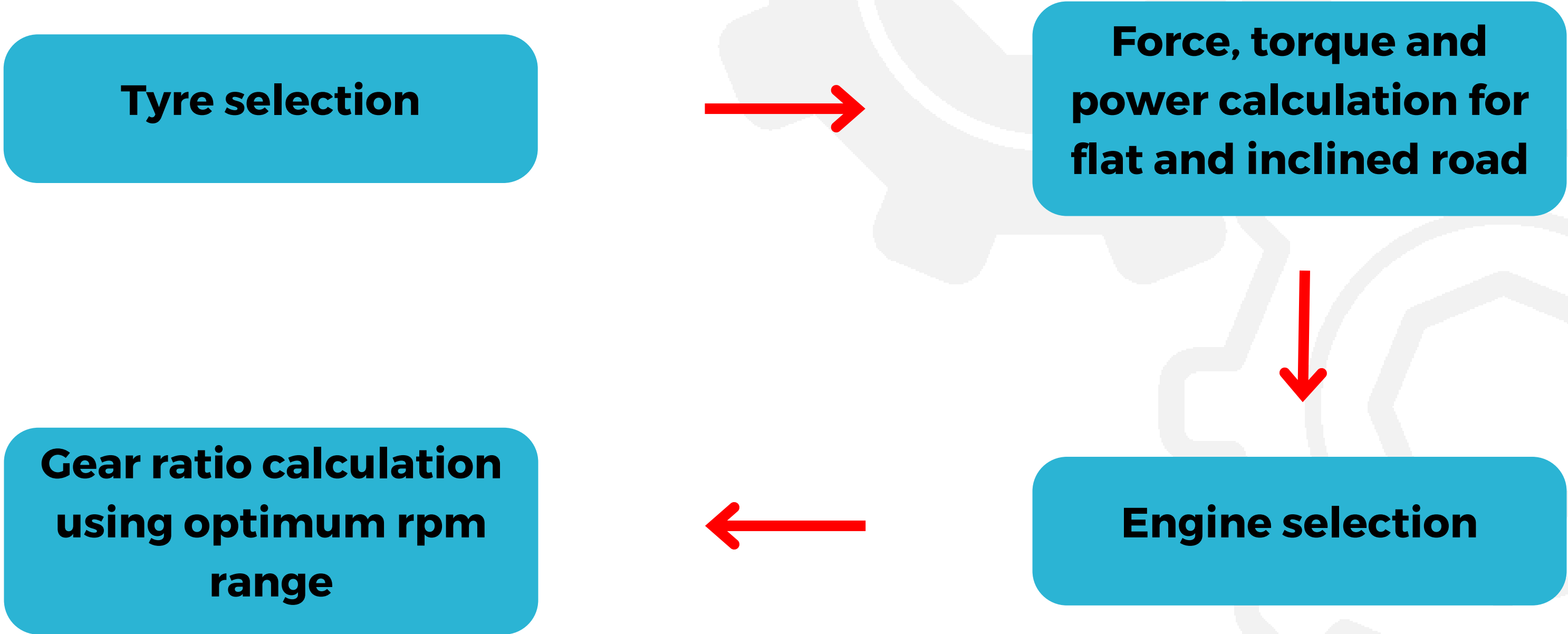
- Maximum speed = 180 kmph
- Drag coefficient = 0.34
- Width = 1930 mm
- Height = 1755 mm



210640
A

Toyota highlander
2024

GEAR RATIO CALCULATION



```
graph TD; A[Tyre selection] --> B[Force, torque and power calculation for flat and inclined road]; B --> C[Engine selection]; C --> D[Gear ratio calculation using optimum rpm range];
```

The diagram illustrates a four-step process for gear ratio calculation, set against a background of interlocking gears. The steps are contained in blue rounded rectangular boxes, connected by red arrows indicating a clockwise flow. The steps are: 1. Tyre selection, 2. Force, torque and power calculation for flat and inclined road, 3. Engine selection, and 4. Gear ratio calculation using optimum rpm range.

Tyre selection

**Force, torque and
power calculation for
flat and inclined road**

**Gear ratio calculation
using optimum rpm
range**

Engine selection

GEAR CALCULATION

**Minimum distance
between shafts**

Teeth count calculation

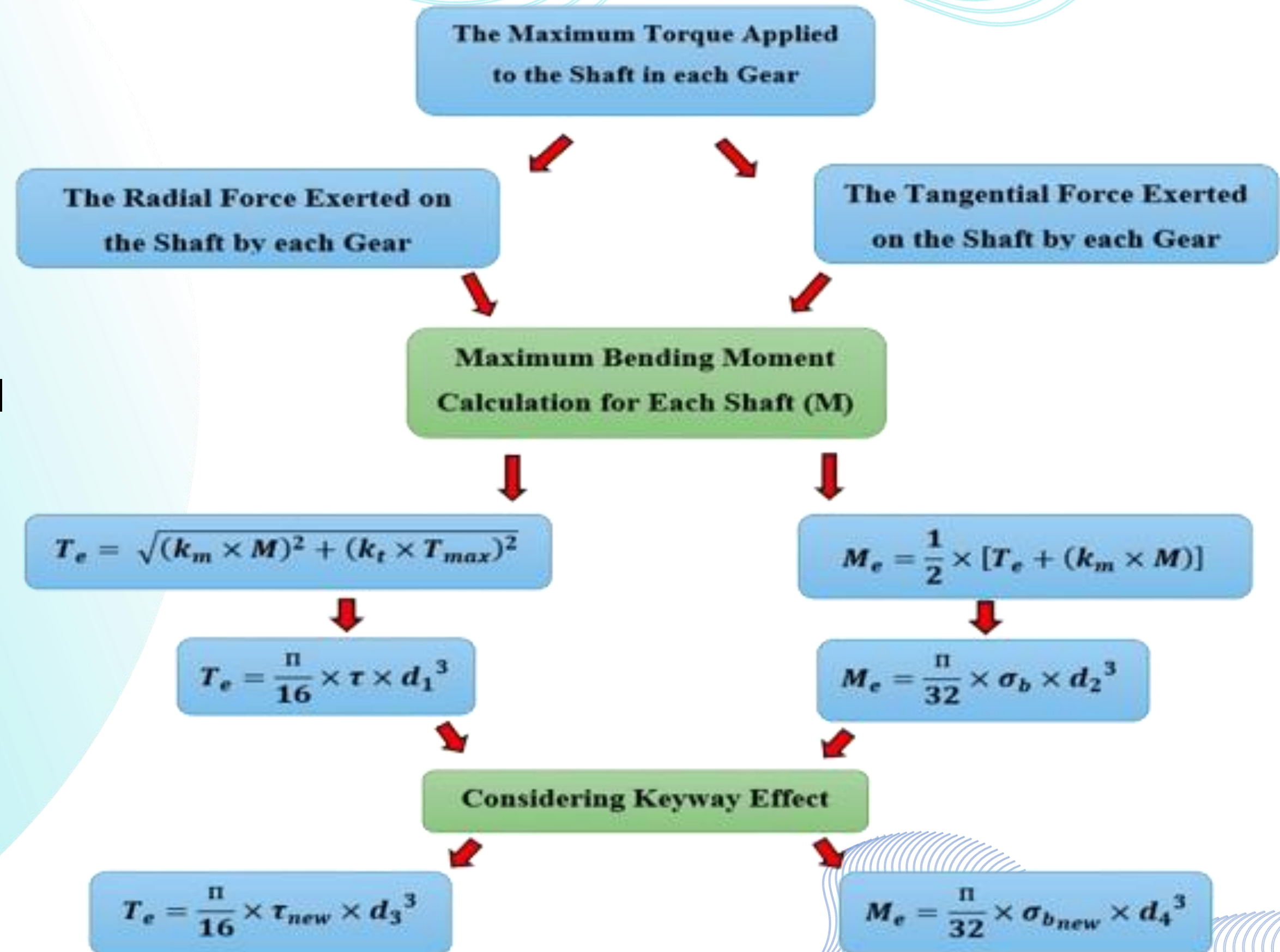
Safety check

**Tangential load,
dynamic load, static
load and wear load
calculation**

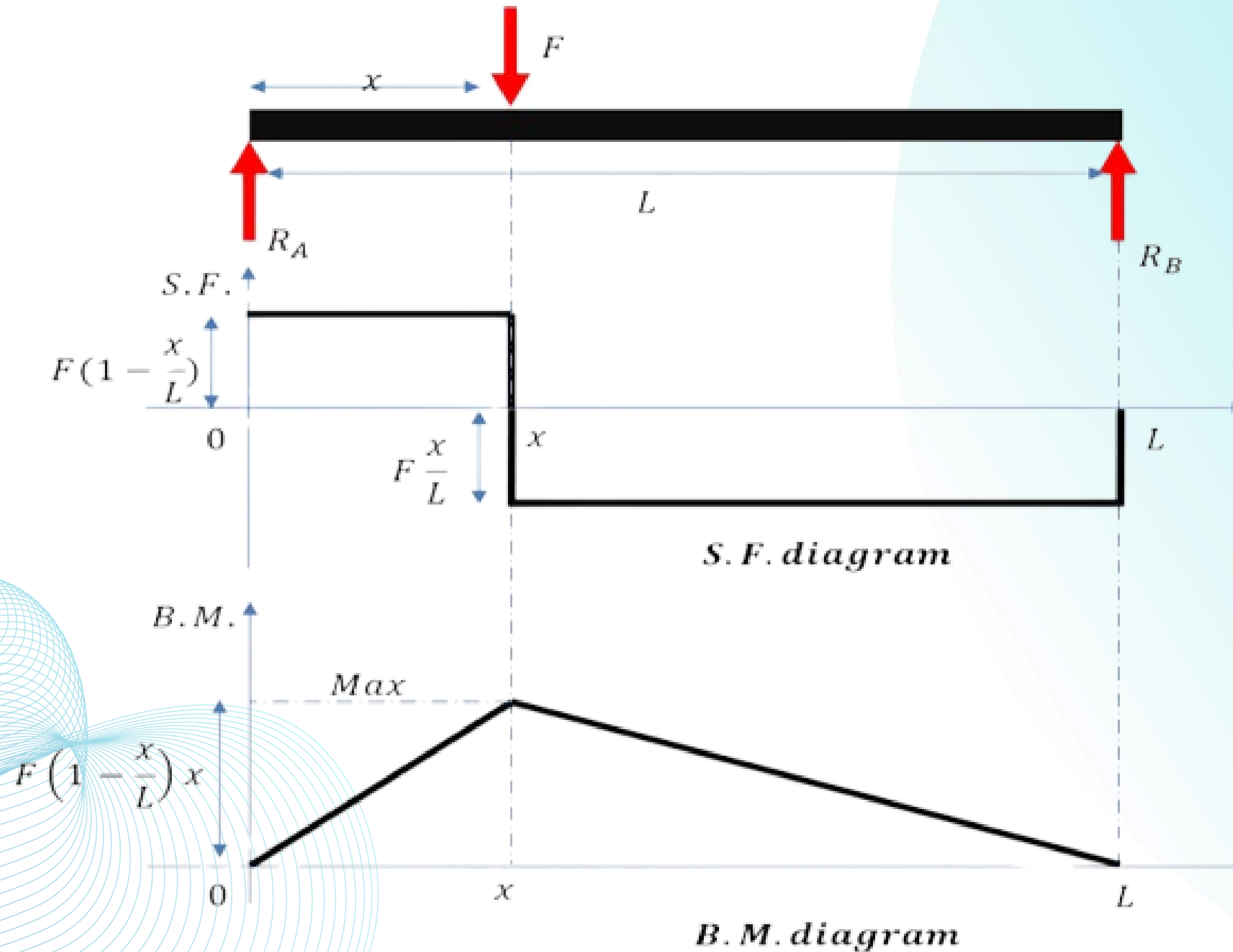
**Material selection for
gear**

SHAFT CALCULATION

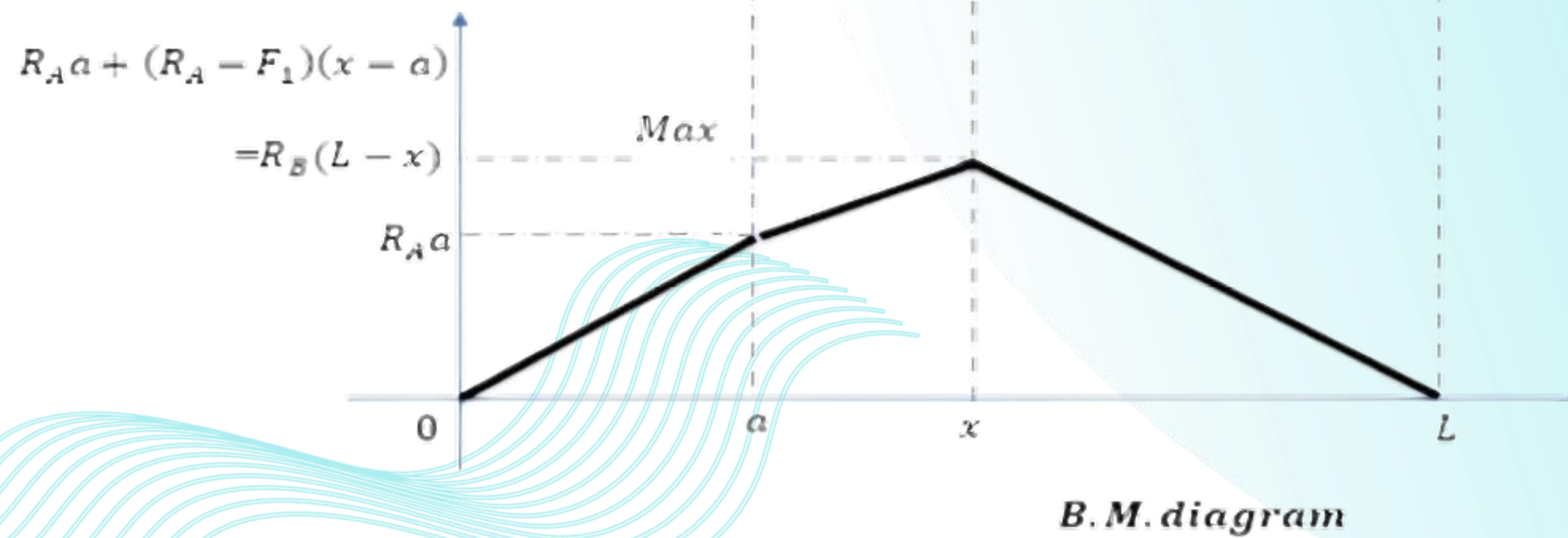
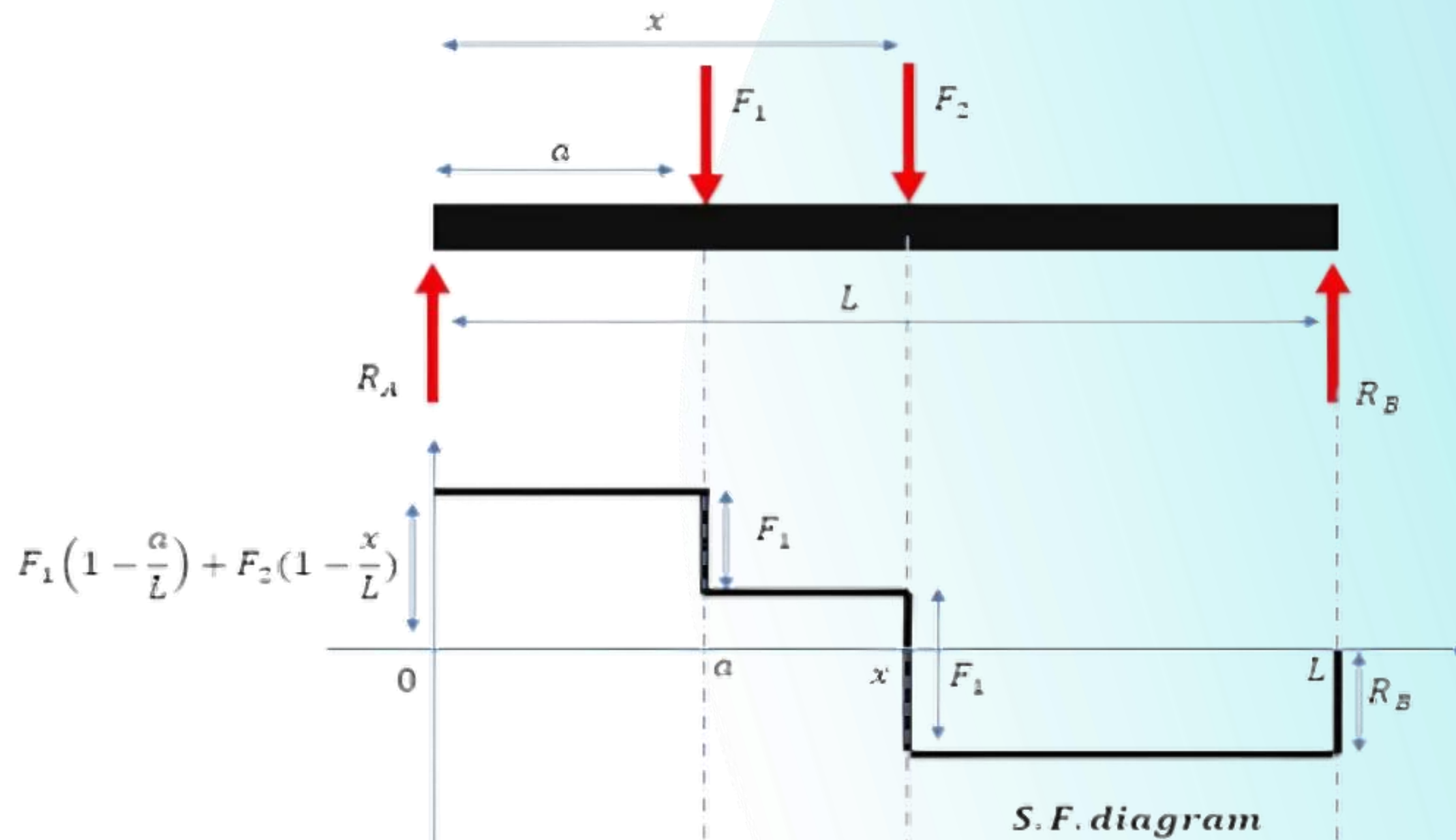
Selected shaft material
AISI 4340 (Steel)



BENDING MOMENT CALCULATION



**For Input shaft, Idler shaft
& Output shaft**



**For Lay
shaft**

Maximum Bending Moment Calculation
for Each Shaft (M_T) at Each Gear for
Tangential Loads

Maximum Bending Moment Calculation
for Each Shaft (M_R) at Each Gear for
Radial Loads

Maximum Bending Moment Calculation for Each
Shaft (M) at Each Gear

$$M = \sqrt{(M_T)^2 + (M_R)^2}$$

BEARING CALCULATIONS

**Final selected material
SAE 52100 (bearing steel)**

$$L_H$$



$$L = 60 \times \omega \times L_H$$



$$W_A = 0$$

$$W_R = \sqrt{(R_{Radial})^2 + (R_{Tangential})^2}$$



$$W_{OR} = (X \times V \times W_R) + (Y \times W_A)$$

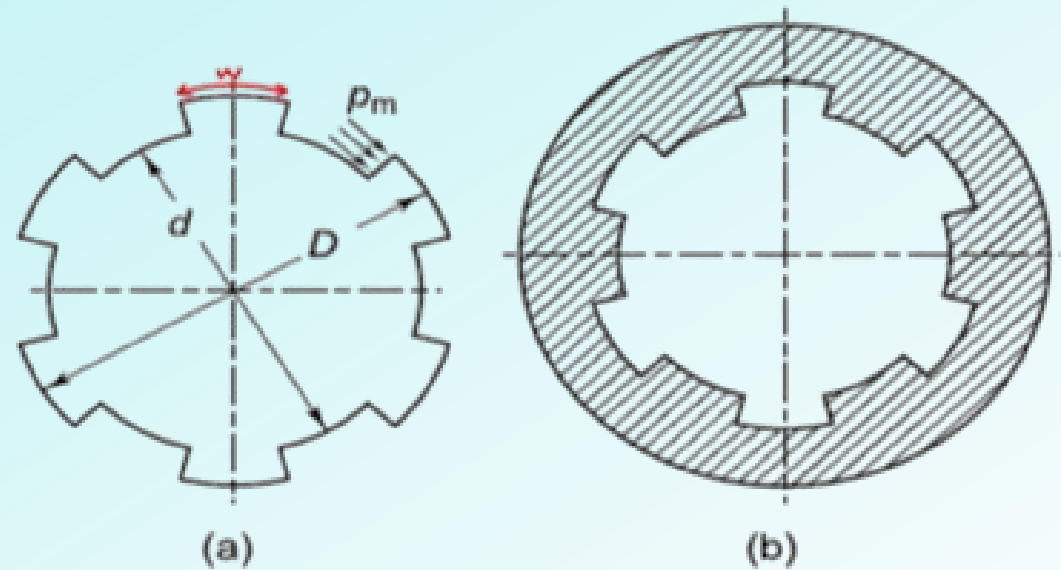


$$W = k_s \times W_{OR}$$



$$L = \left(\frac{C}{W}\right)^k \times 10^6$$

SPLINE CALCULATION



Splines: (a) Shaft (b) Hub

$$T_{max} = \frac{1}{8} \times \tau \times n \times l \times (D^2 - d^2)$$

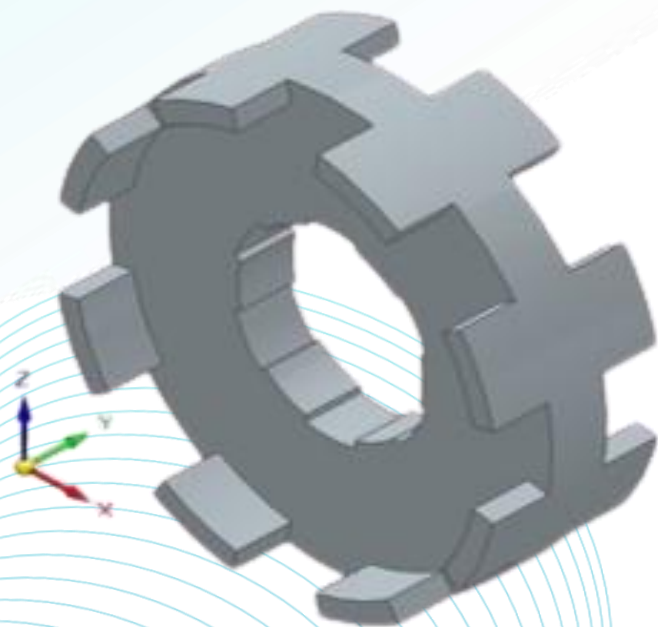
“d” value calculation under constant “D”

Then $d >$ output shaft minimum diameter



“D” value calculation under constant “d”

$$W = D \times 0.25$$



KEY CALCULATIONS

Determining width and thickness



Calculating length



Keyway effect calculation



Safety check

**210494
D**

LUBRICATION SYSTEM

Purpose

- Reduces friction and wear between parts.
- Controls heat and prevents corrosion.

Lubricant Selection

- SAE 75W-90 Gear Oil: High durability and corrosion resistance.
- Use synthetic oils for extreme temperature stability.

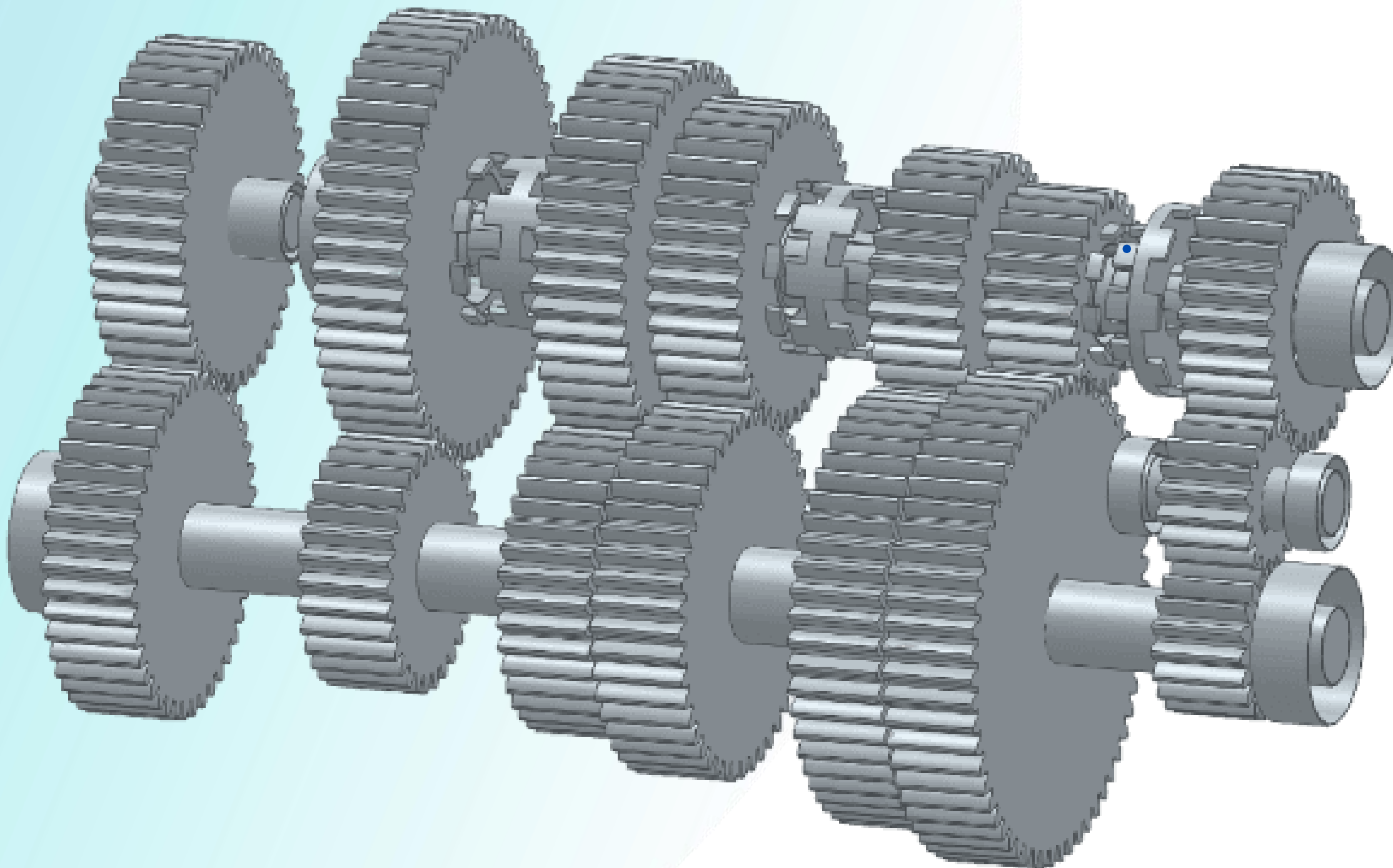
Methods

- Splash Lubrication:
Gears rotate and splash oil for continuous lubrication.
- Pressure Lubrication:
A pump circulates oil to critical, high-load areas.

Maintenance Schedule

- Check oil levels every 5,000–10,000 km.
- Replace oil every 30,000–40,000 km.

FINAL DESIGN



Selected materials

- Gears: AISI 8620 steel (oil-quenched for durability).
- Shafts: AISI 4340 steel (high tensile strength).
- Bearings: SAE 52100 bearing steel (wear resistant).
- Keys: AISI 6510 provides better reliability under load

**THANK
YOU**